



Prof. dr. Rashmi Sasidharan

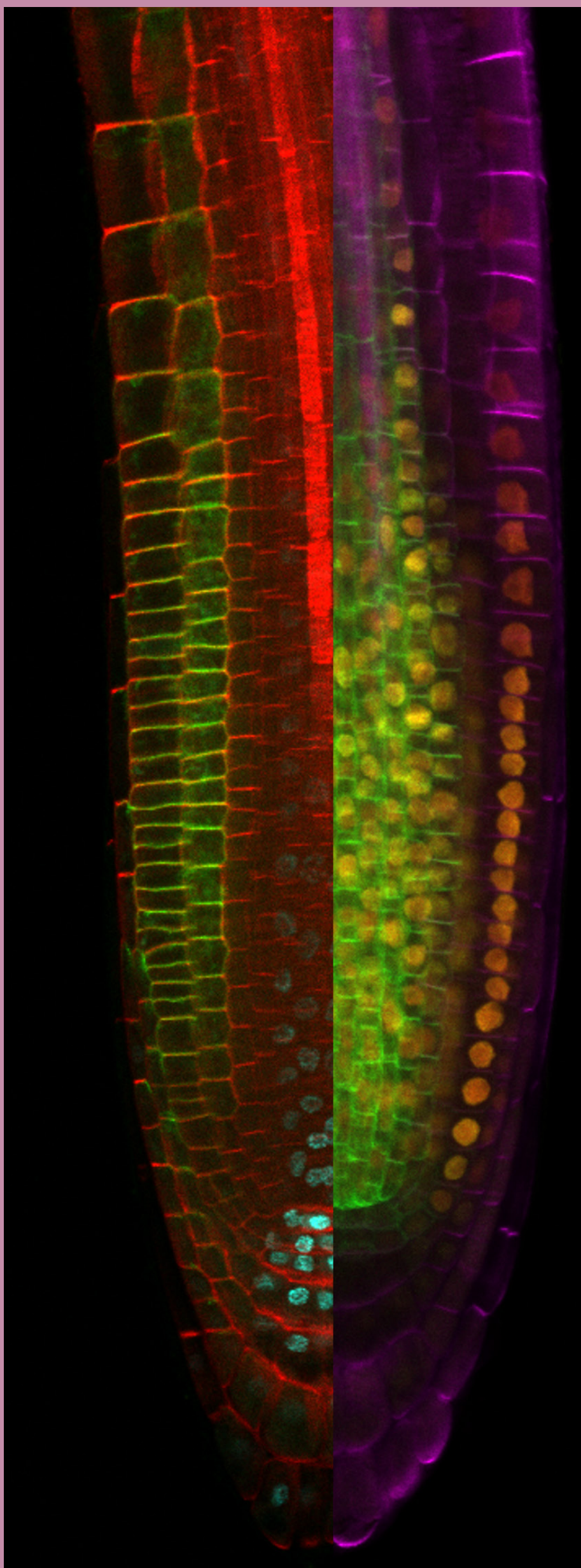
Nowadays I wear two hats. I have my own research line which has been primarily on understanding plant responses to water stress on both extremes (flooding and drought). And in the chair group, we’re also focused on temperature, salinity, and multistress resilience. We work with multiple species, from *Arabidopsis* to several crops.

ON THE IMPACT

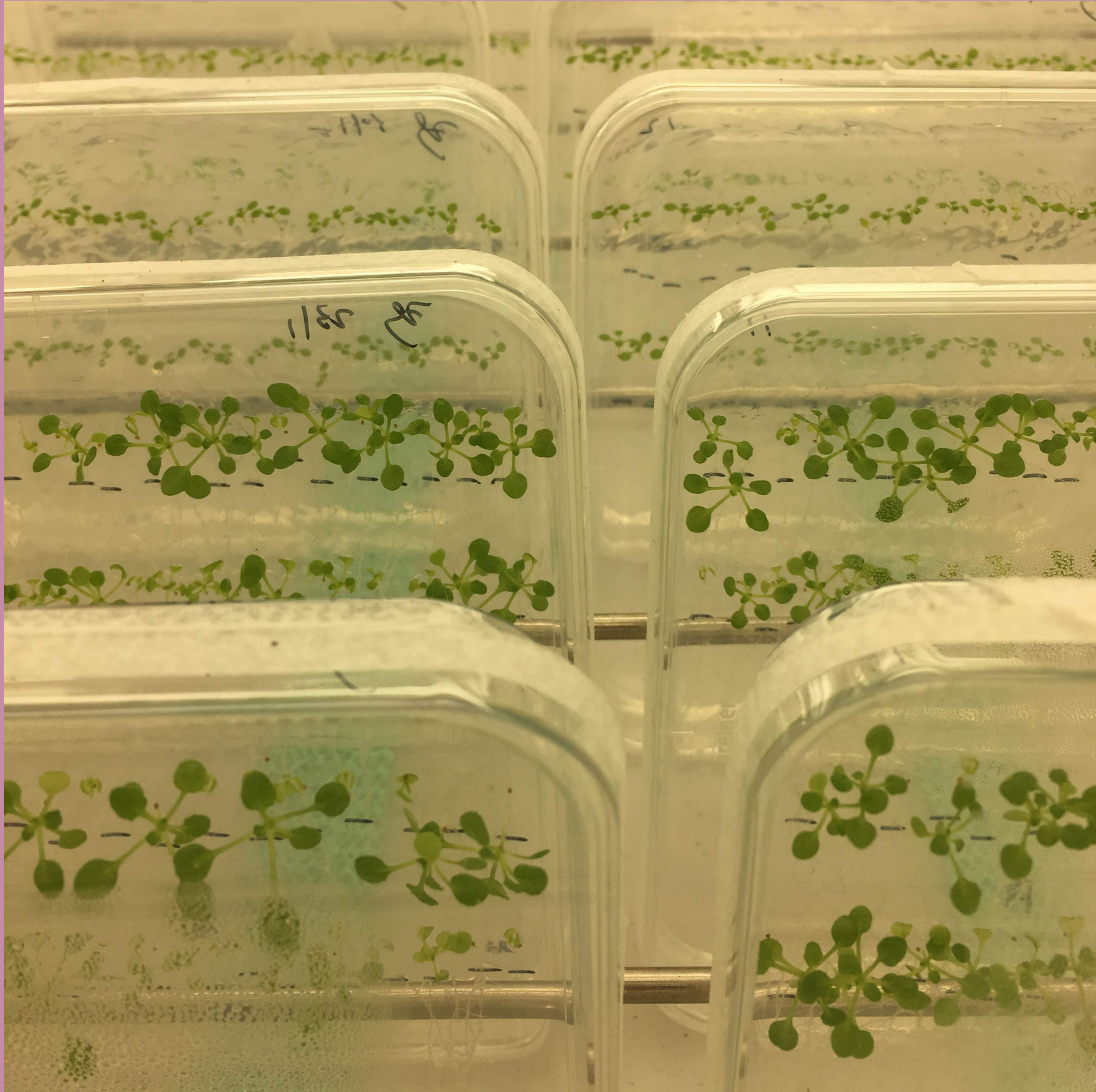
We are not necessarily moving towards breeding a specific species and giving that to the farmer. But we do believe the knowledge we are generating will contribute to that step, which probably is done by a breeding institute or a company.

Through education, we are also making a difference. We impart knowledge and influence opinions through training young people to think based on scientific facts.

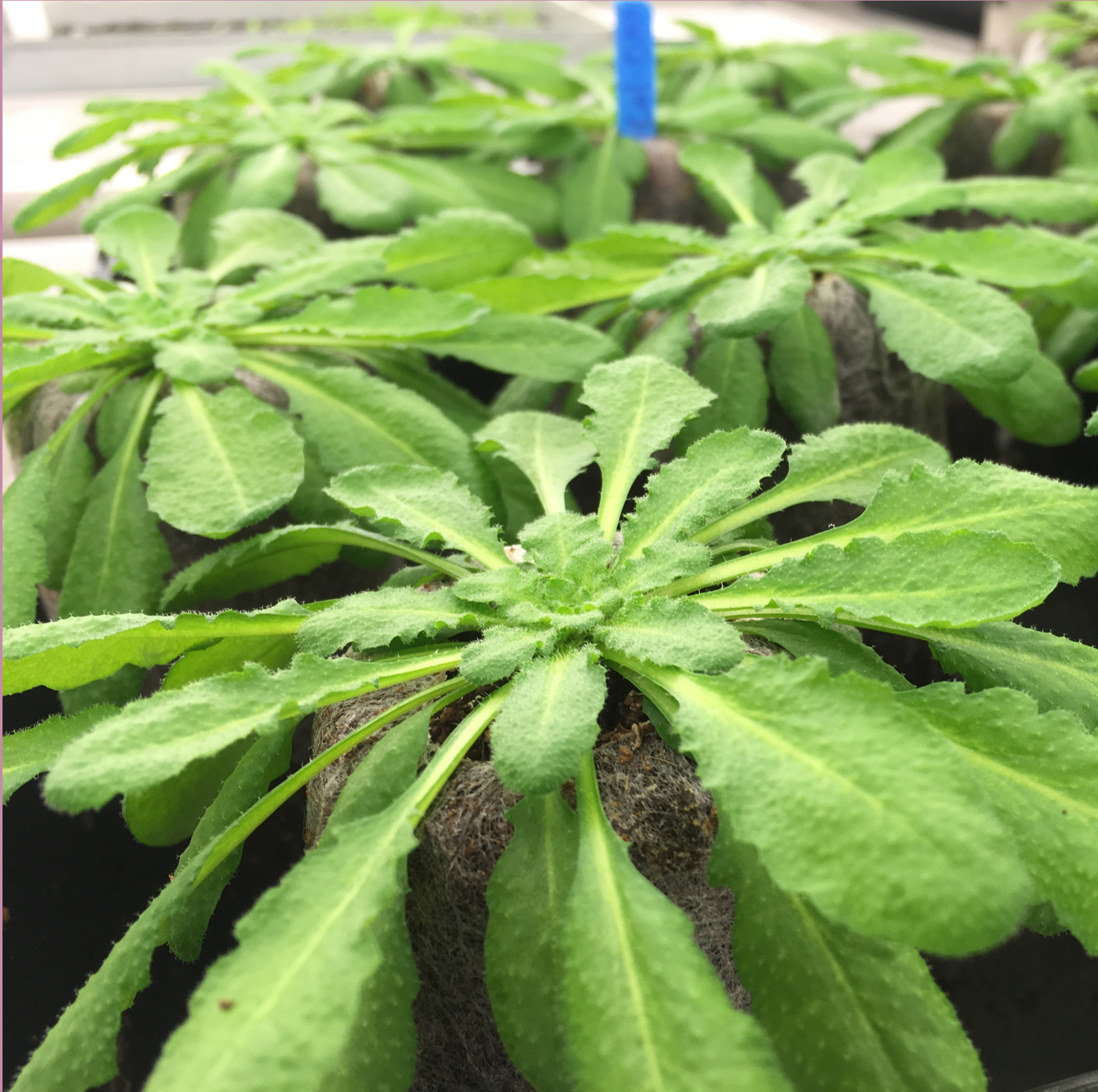
Who is responsible for communicating science to the public?



*Arabidopsis* roots marked with different proteins seen under a microscope (by Gyöngyi Macías Honti)



Young *Arabidopsis* plants growing on plates with nutrients. Their roots are studied for their response to environmental stress.



*Arabidopsis* plants growing on soil. They help understand plant’s responses to environmental stress.

ON PREVIOUS EXPERIENCES

I find live interaction more rewarding. People’s responses are interesting when I talk about how cool plants are because they are extremely flexible to stress. Seeing that wonder on people’s faces, that’s cool. Then, when you speak to a non-plant scientist pool of people, I realize that the same thing which has become so mundane to me is not for them.

ON THE CHALLENGES

I think we should focus more on science communication. It is outside of our comfort zones to try and explain our research in a fun and easy way to people beyond our immediate academic sphere, so I think we naturally tend to avoid it. Currently there’s no excuse for why we shouldn’t do it. There’s plenty of opportunities. The university provides plenty of training. We just have to make time and go do it.

If you ask most people if plants are important, they will say yes. However, research investment in plant sciences is like a drop compared to the ocean going to medical sciences. Why? Because of the perception amongst the government and policy makers that one domain is more important than the other.

As scientists, we have a big responsibility to speak out about how important it is to invest in new technologies that make us less reliant on fertilizers and pesticides. The GMO debate has ensured that the public is very hesitant to new technologies. This hesitation causes breeding companies to not use these technologies. When they don’t use them, the funding for their research goes down. If we all sit in our little lab towers and we don’t participate, it will negatively affect us.

ON THE FUTURE

I think there are a few things we can do as scientists for science communication:

If someone reaches out to you with a science communication opportunity, which is happening more often, take it.

Increase the awareness of the fact that there is plant science research being done, that it is an active field, you can have a career out of it, and how it has contributed or is contributing to society.

Connect with other disciplines, like social sciences. This can help us understand what we can do to help society accept new technologies.



Potato plant grown in soil. Knowledge derived from *Arabidopsis* plants is applied on crops like potato to understand their response to environmental stress.

